Exocrine pancreas, liver & biliary system

Medical physiology

ILOs

1. Describe the composition and functions of exocrine pancreas.
2. Describe control of pancreatic secretion.
3. Know the functions of liver.
4. Name the composition, functions & control of bile secretion.
5. Know the functions of gall bladder and classify the various types of jaundice

Pancreatic secretion

Volume: 1500 ml/day

Alkaline (due to high bicarbonate content: 113 meq versus 24 meq in plasma)

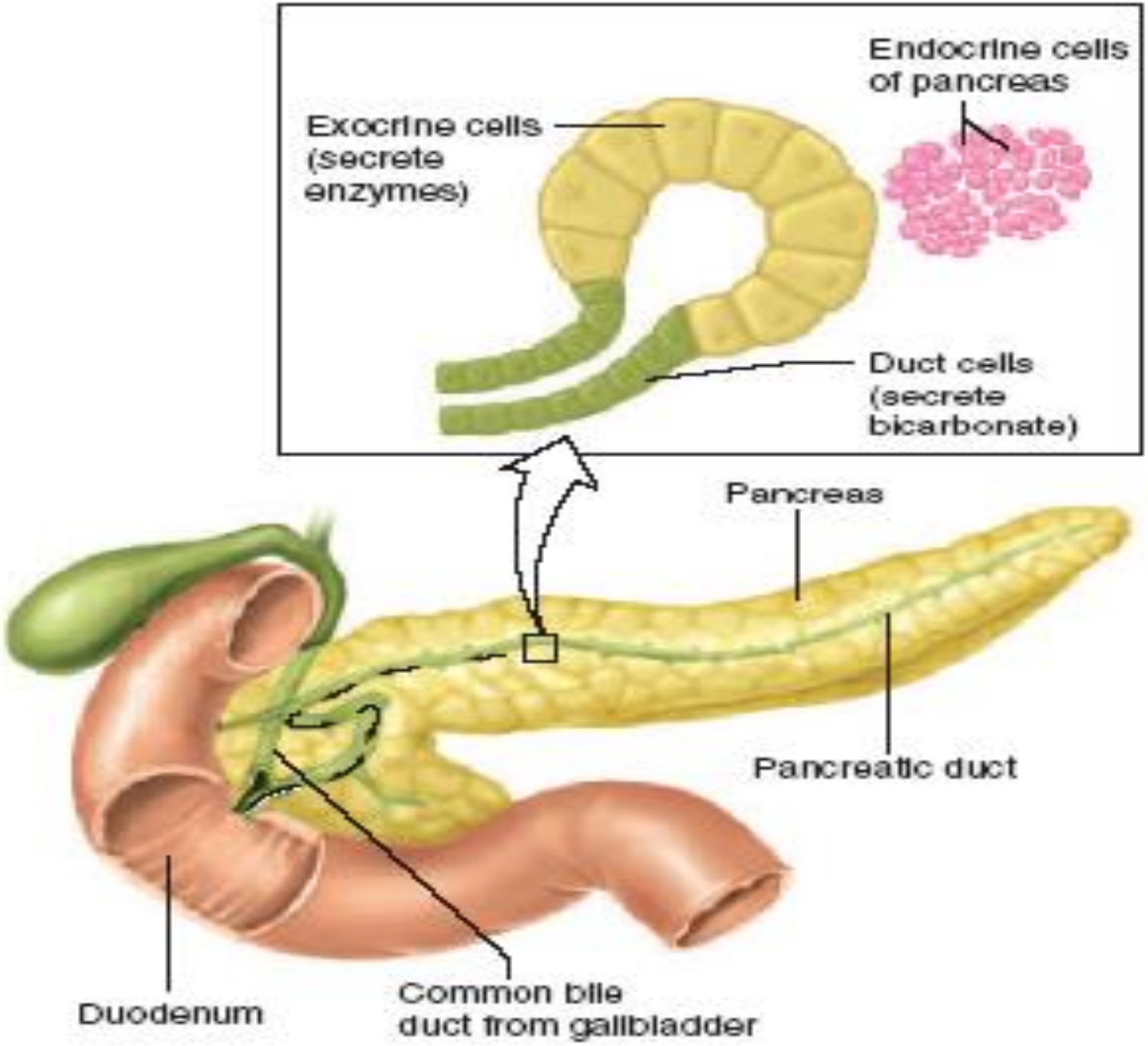
Pancreatic, bile & intestinal juices neutralize Hcl raising pH of duodenum to 6-7.

So, jejunum contents (i.e. intestinal contents) are neutral & rarely alkaline

Composition: Water + bicarbonate + enzymes

Pancreatic enzymes: are fully sufficient for complete digestion

even in absence of salivary amylase & gastric pepsin


Endocrine cells
of pancreas
Exocrine cells
(secrete
enzymes)
Duct cells
(secrete
bicarbonate)
Pancreas
Pancreatic duct
Duodenum
Common bile
duct from gallbladder

- **Mechanism of secretion of aqueous (bicarbonate) secretion:**

1) In duct cells:

- $CO_2 + H_2O \xrightarrow{\text{carbonic anhydrase}} H_2CO_3 \longrightarrow H + HCO_3$

HCO_3 is actively secreted in the lumen by HCO_3 ATPase

2) H^+ ions are pumped out to the plasma in exchange with Na^+ ions into the duct cells

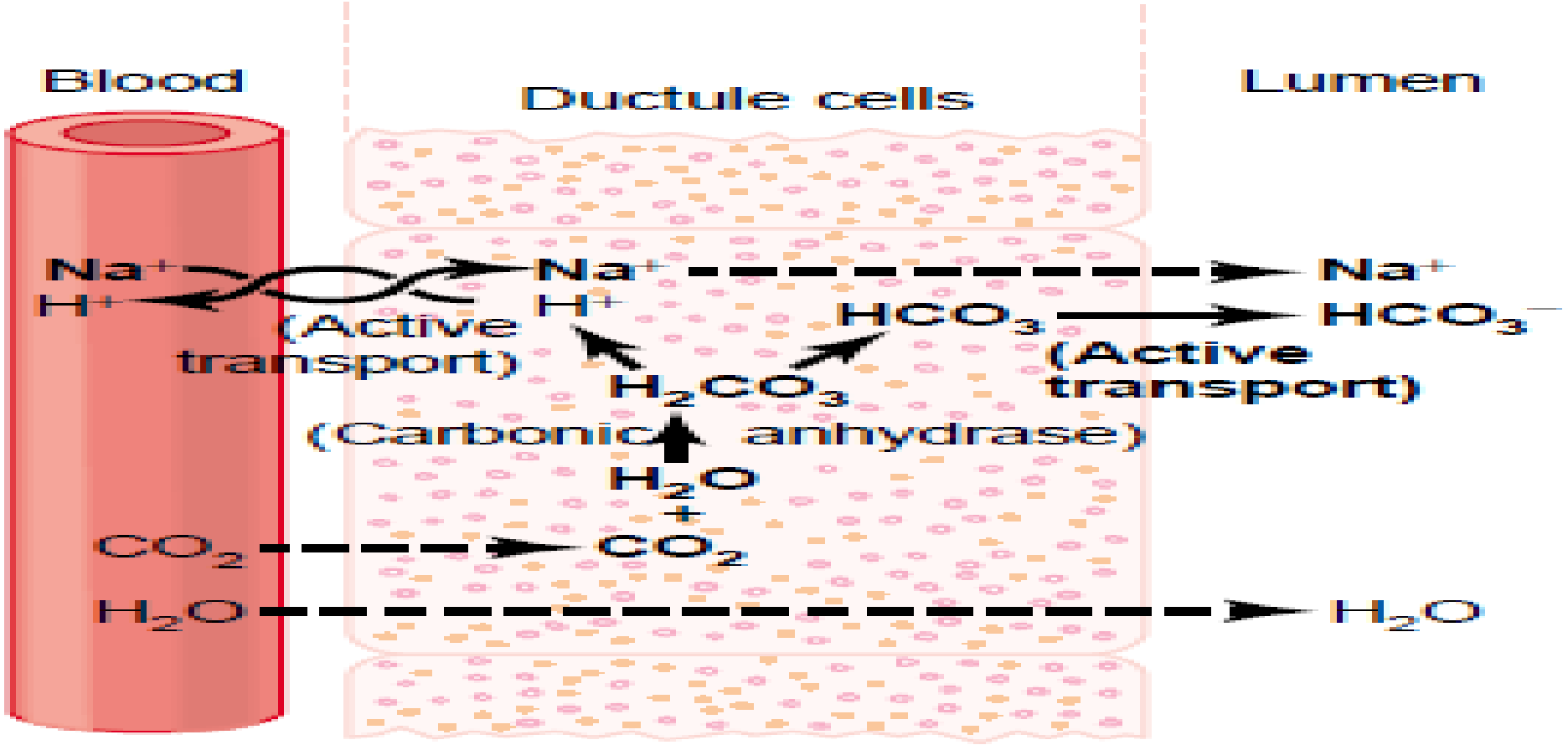
Then Na^+ ions are then transported by diffusion or active transport into lumen

3) H^+ entering the blood unites with HCO_3 to form carbonic acid which dissociates to CO_2 & H_2O.

Then CO_2 diffuses to inside of duct cell & repeat the cycle

3) Na^+ & HCO_3 creates an osmotic gradient that causes osmosis of water

NB: The Acid tide: it is the decrease in pH of pancreatic venous blood caused by H^+ actively pumped to blood. It neutralizes the alkaline tide of gastric venous blood



- **Functions of aqueous secretion:**

1-It washes out the enzymes which are secreted by acini

2-It provides a neutral pH for enzymatic activity in the intestine

Regulation of Pancreatic secretion

Under nervous & hormonal control **but** *hormonal regulation is the main mechanism of regulation*

1- Nervous control:

During cephalic & Gastric phase: Of gastric secretion:

There is stimulation of vagus (unconditioned & Conditioned reflexes) resulting in:

Acetylcholine release which acts on the **acinar cells** to secrete the **enzymatic** juice

2- Hormonal control:

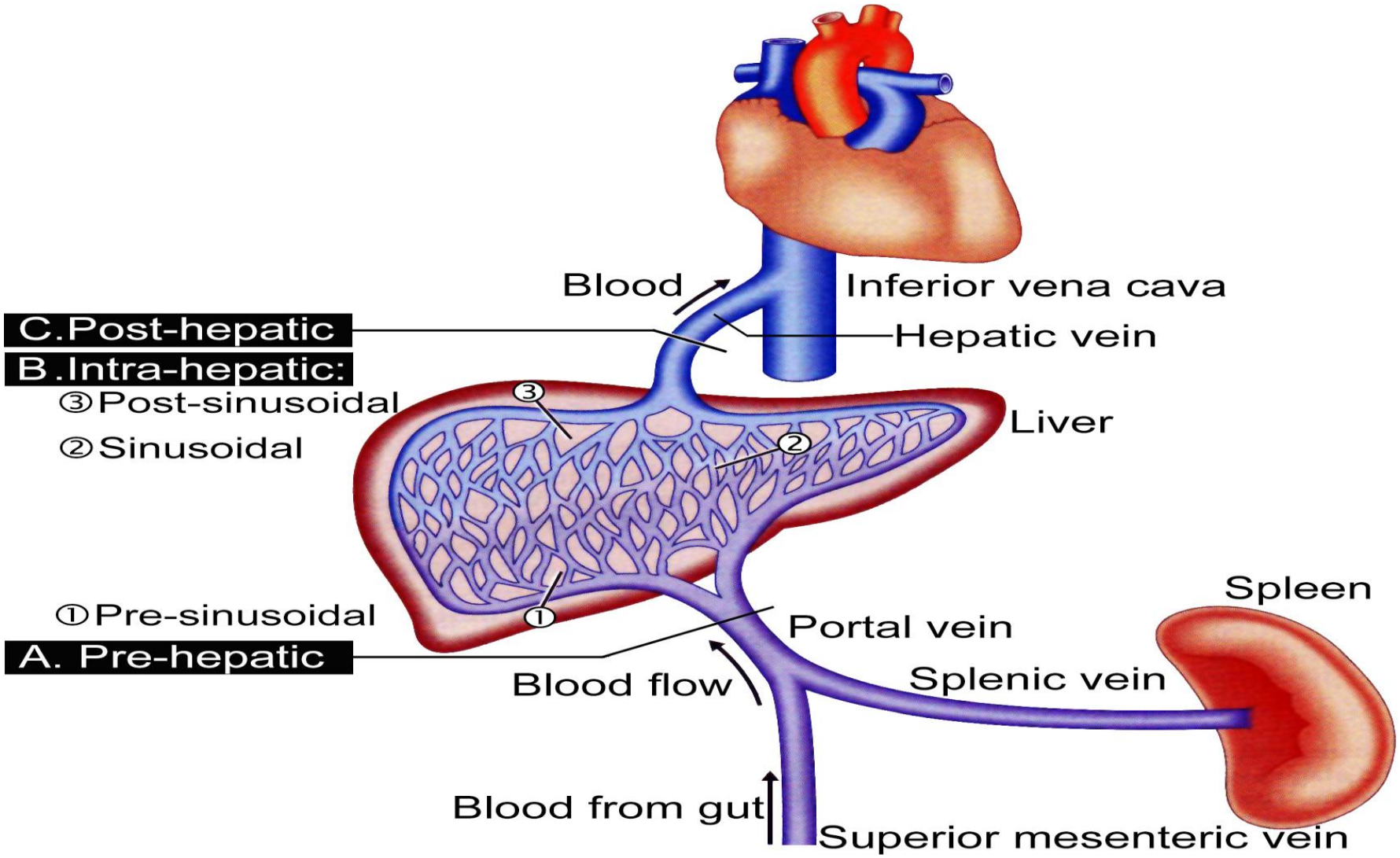
1. Secretin:

- It is **secreted from** upper small intestine
- Its **release is stimulated by**: Acid chyme in intestine
- It **acts on pancreatic duct** to stimulate **alkaline pancreatic juice** which neutralize acid chyme from stomach to provide optimum pH for action of pancreatic enzymes
- It is **mediated by**: cAMP

2. Cholecystokinin "CCK" pancreozymin:

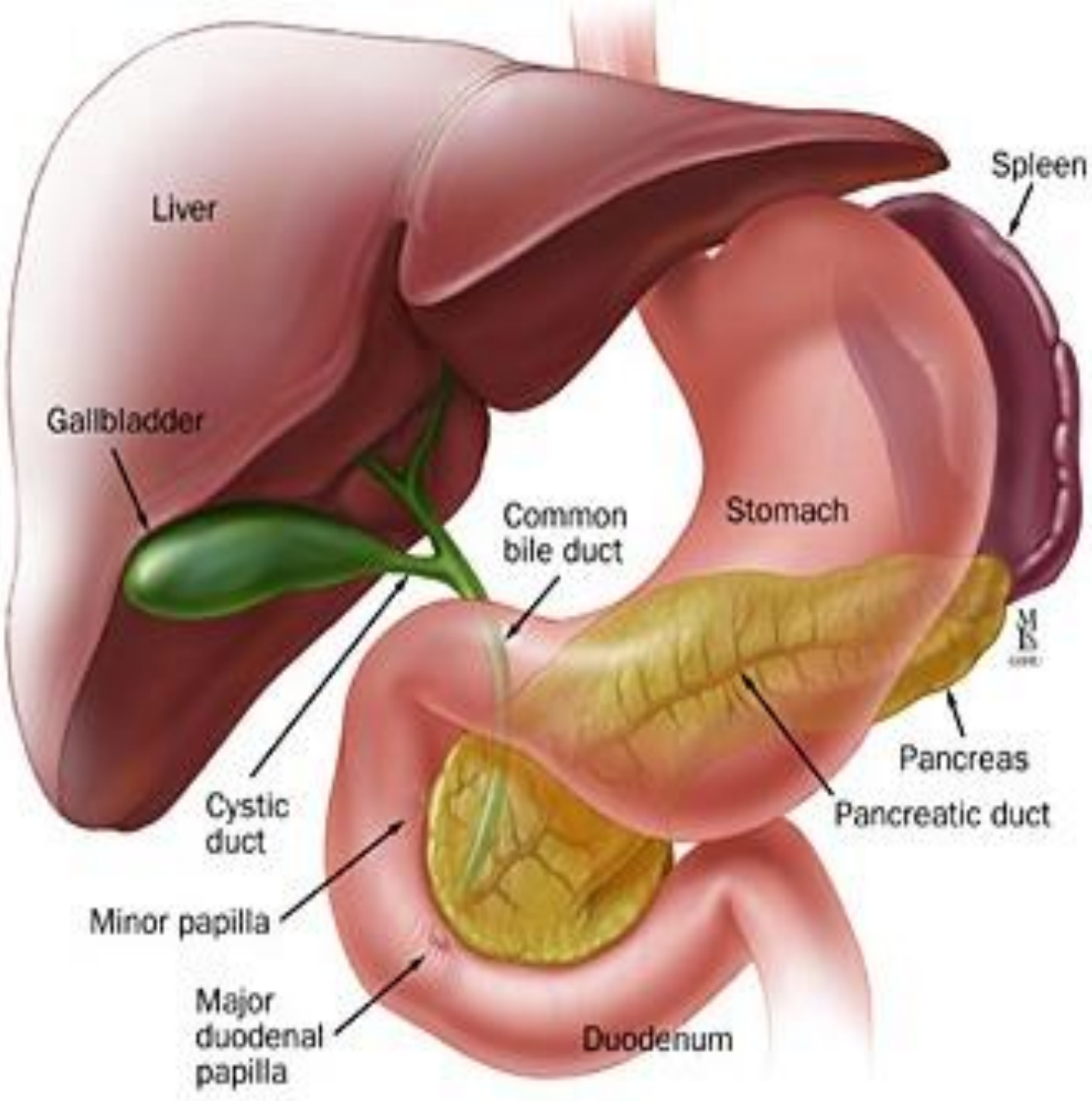
- It is **secreted from** upper small intestine
- Its **release is stimulated by**:
  ① Polypeptides
  ② Amino acids
  ③ Fatty acids
  ④ Bile salt
- It **acts on pancreatic acini** to stimulate pancreatic juice rich in enzymes
- It is **mediated by**: Phospholipase C

Liver and biliary system



❶ Liver:

- The functional unit of the liver is: **Liver lobule**:
  - Is composed of hepatocellular plates radiating from the central vein
  - Central vein empties into hepatic veins
- In between the cells lie small bile canaliculated that empty into bile ducts that lie in the fibrous septa separating liver lobules


Liver
Spleen
Gallbladder
Common
bile duct
Stomach
Pancreas
Pancreatic duct
Cystic
duct
Minor papilla
Major
duodenal
papilla
Duodenum

❷ Gall bladder:

In normal individual, bile flows into the gall bladder when sphincter of Oddi is closed

★ **Functions of gall bladder:**

① Storage of bile: Until needed in duodenum

② Concentration of bile and equalization of pressure:

- **Liver bile**: One liter/day while max volume of gall bladder is 20 - 60 ml
- **Mechanism**
  - Na^+ ions are actively transported while Cl^- & HCO_3 ions follow passively
  - Water follows passively by osmosis (H_2O content is 89 %)
  - Concentration of cholesterol, bile salts, lecithin & bilirubin increases

③ It prevents marked rise in biliary Pressure:

Biliary pressure increases to 320 mmHg if cystic duct clamped

④ Acidification of bile:

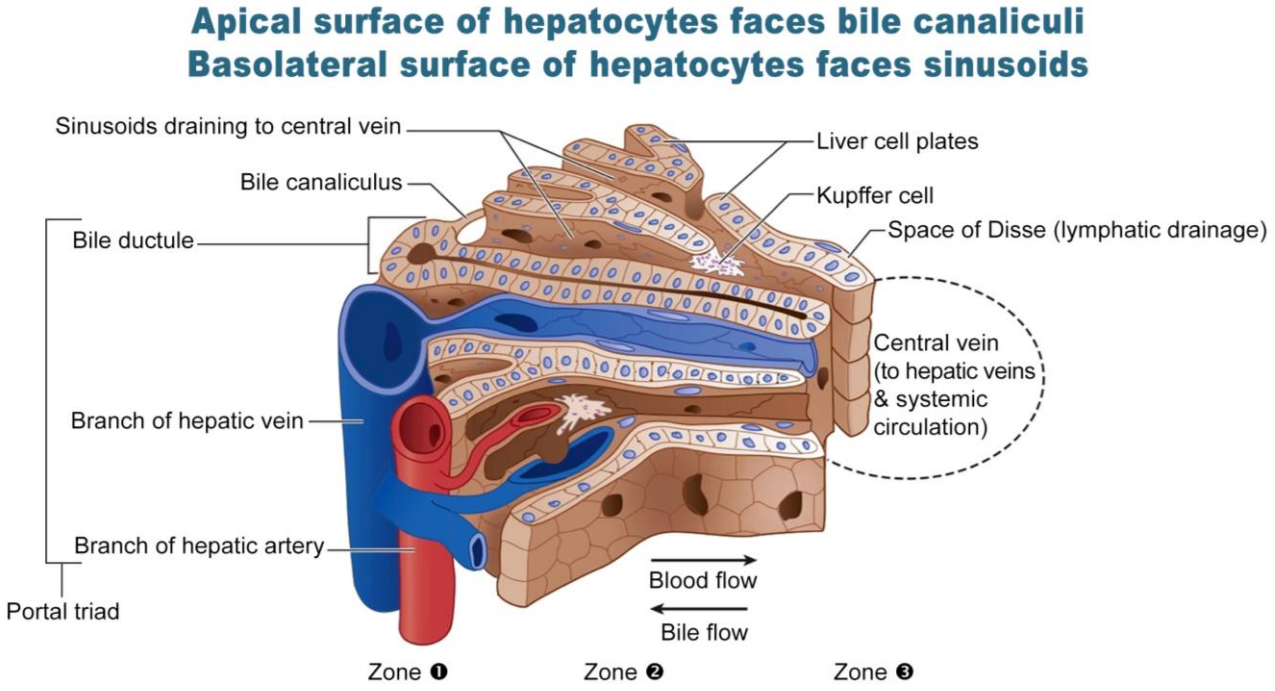
- **Mechanism**: $NaHCO_3$ is reabsorbed, so liver contains 28 mEq/L HCO_3 while gall bladder contains 10 mEq/L
- This mechanism keeps lecithin, cholesterol & bile salts in solution

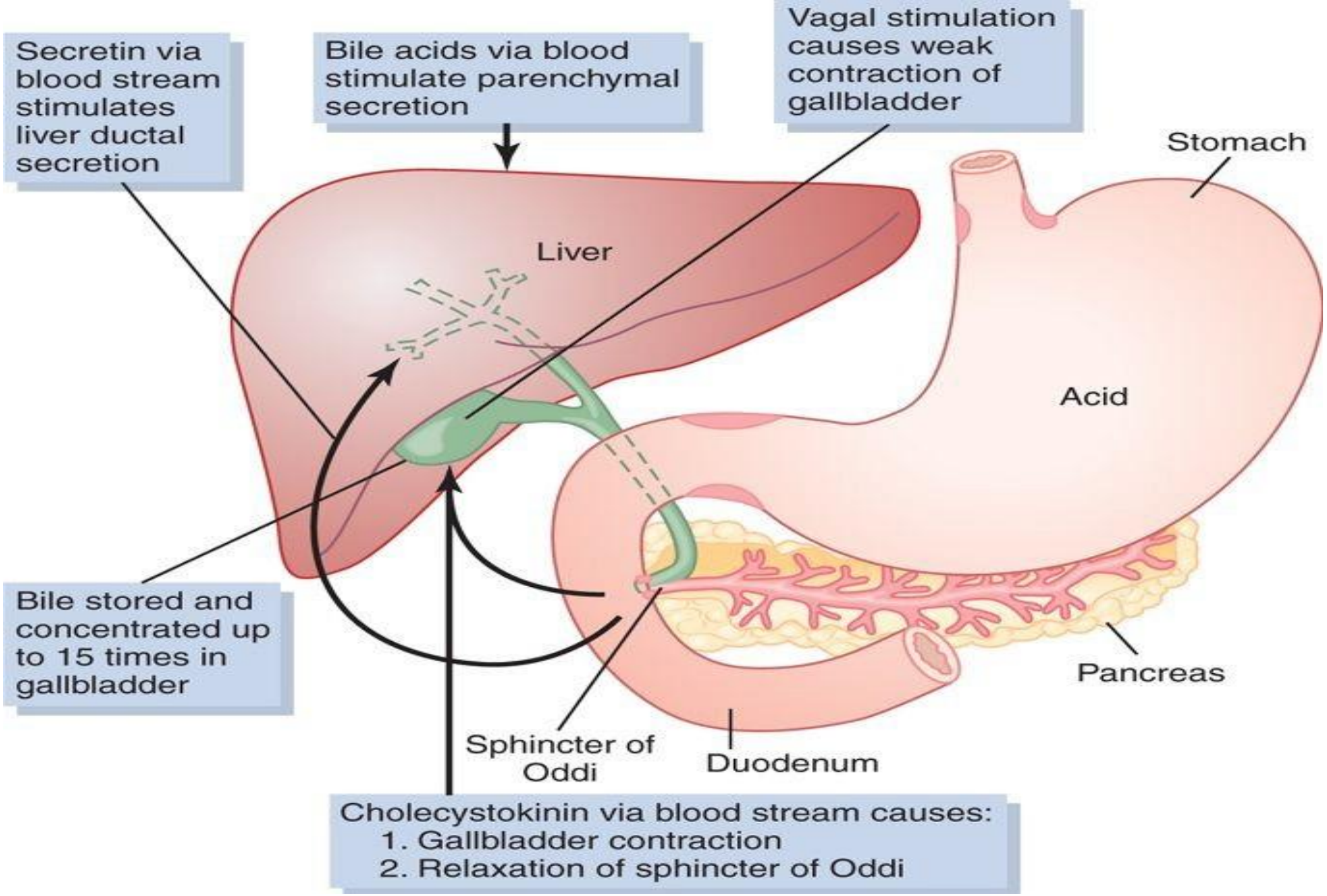
⑤ Secretion of white bile (mucus):

This prevents irritation of the mucosa by the acidified concentrated bile

⑥ Evacuation of gall bladder:

- **Control**
  - Hormonal: CCK
  - Nervous: Vagal stimulation directly during cephalic phase and indirectly through vagovagal reflex
- Vagal stimulation causes a less strong gall bladder contraction and sphincter of Oddi relaxation than CCK




Secretin via blood stream stimulates liver ductal secretion
Bile acids via blood stimulate parenchymal secretion
Vagal stimulation causes weak contraction of gallbladder
Stomach
Liver
Acid
Bile stored and concentrated up to 15 times in gallbladder
Pancreas
Sphincter of Oddi
Duodenum
Cholecystokinin via blood stream causes:
1. Gallbladder contraction
2. Relaxation of sphincter of Oddi

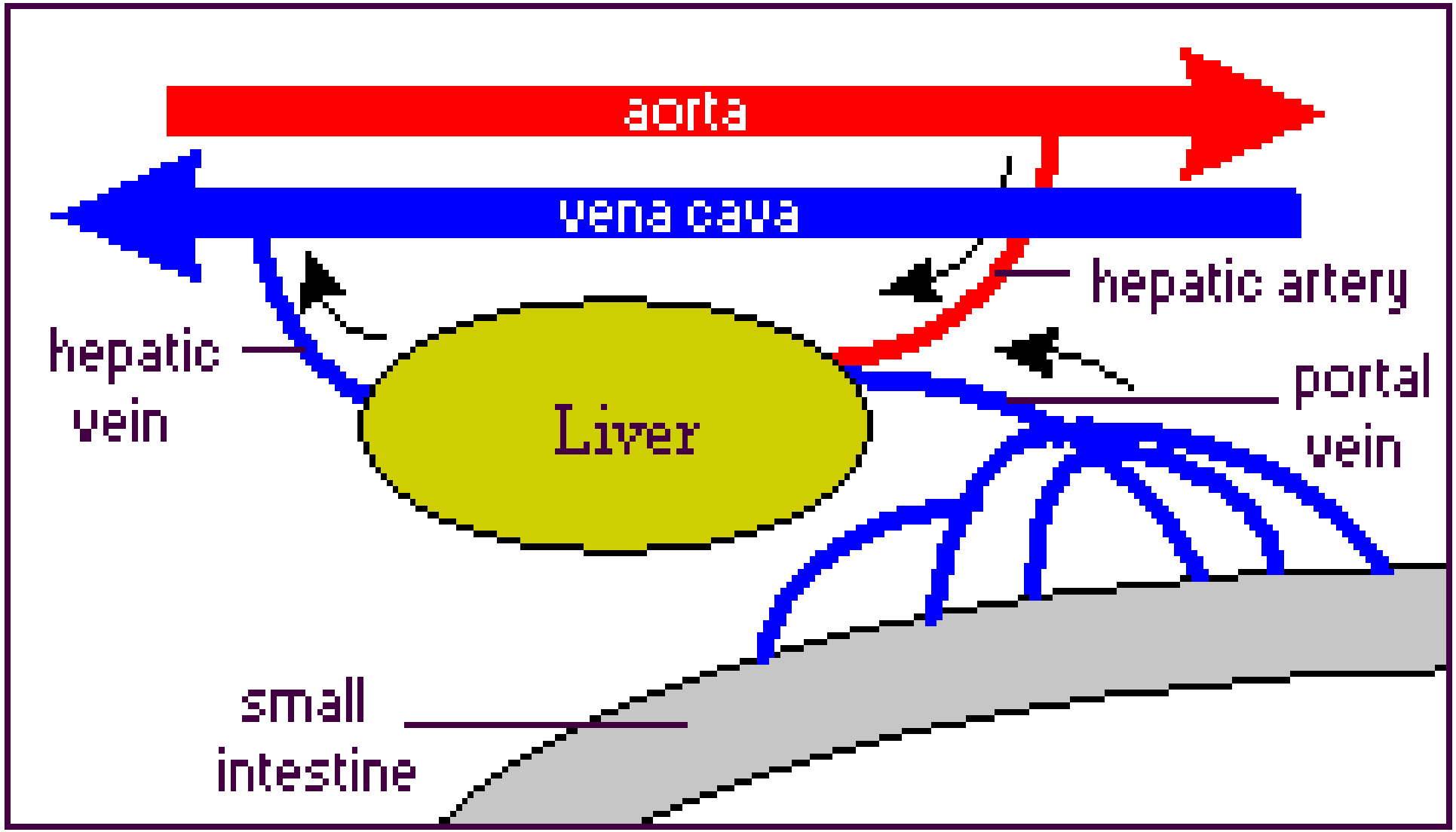
Liver functions:

① Vascular functions *(For storage & Filtration of blood)*

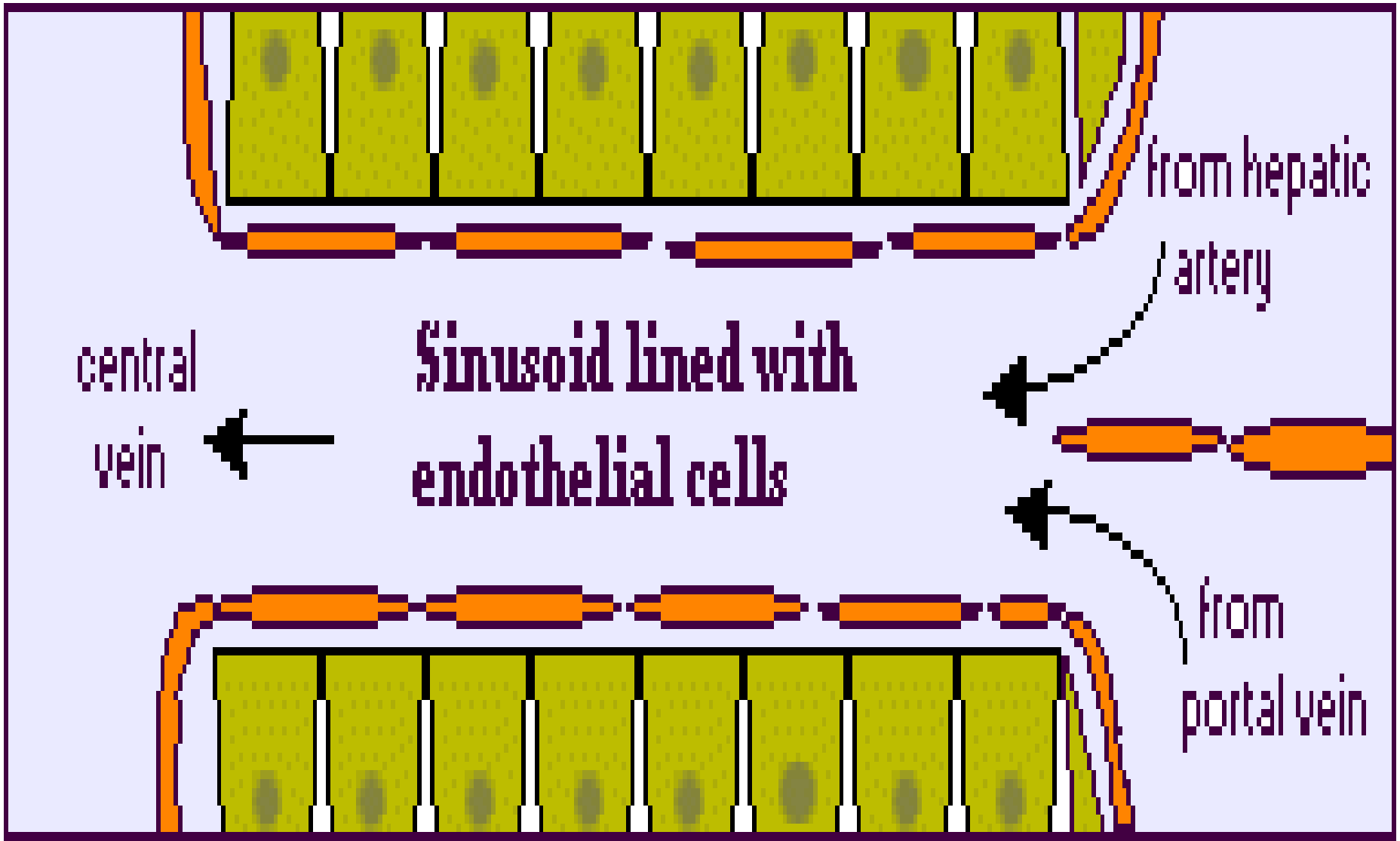
1. The liver can store:
   - 200 - 400 ml of blood
   - Which can replace the blood lost in hemorrhage

2. The liver sinusoids are lined by many VonKupffer cells (macrophages) which remove 99% of bacteria in the portal blood

The Hepatic Vascular System



Hepatic venous drainage



② Metabolic functions:

1. Carbohydrate metabolism:

The liver plays an important role in glucostatic mechanism via:

- Glycogenesis
- Glycogenolysis
- Gluconeogenesis
- Conversion of galactose & fructose to glucose

2. Fat metabolism:

- Oxidation of fatty acids to supply energy
- Formation of:
  - Very low density lipoprotein (VLDL)
  - Low density lipoprotein (LDL)
  - High density lipoprotein (HDL)
  - Cholesterol, Phospholipids

- Lipogenesis: Conversion of:
  - Carbohydrates to fat
  - Proteins to fat

3. Protein metabolism:

a) Deamination of amino acids before transformation to carbohydrate or fat

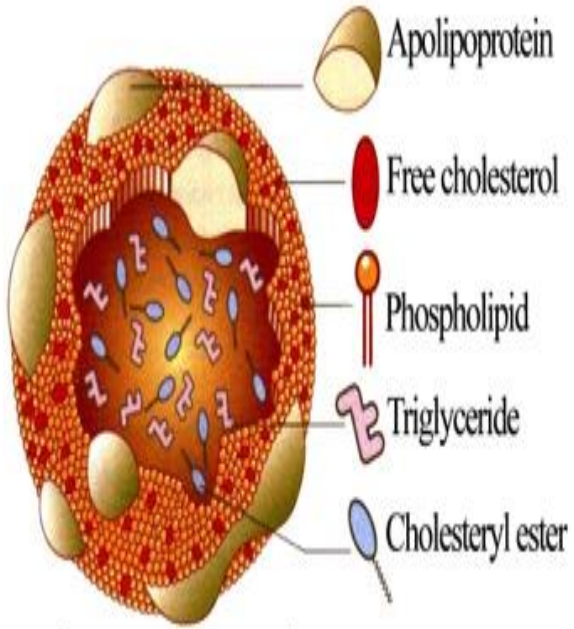
b) Formation of:
1. Urea
2. Most plasma proteins: As albumin
3. Coagulation factors

c) Synthesis of all non-essential amino acids

d) Storage of vitamins: Such as vitamin A, D & B12

e) Storage of iron: Iron combines with apoferritin to form ferritin

f) Detoxification [or excretion in bile]:



Structure of lipoproteins

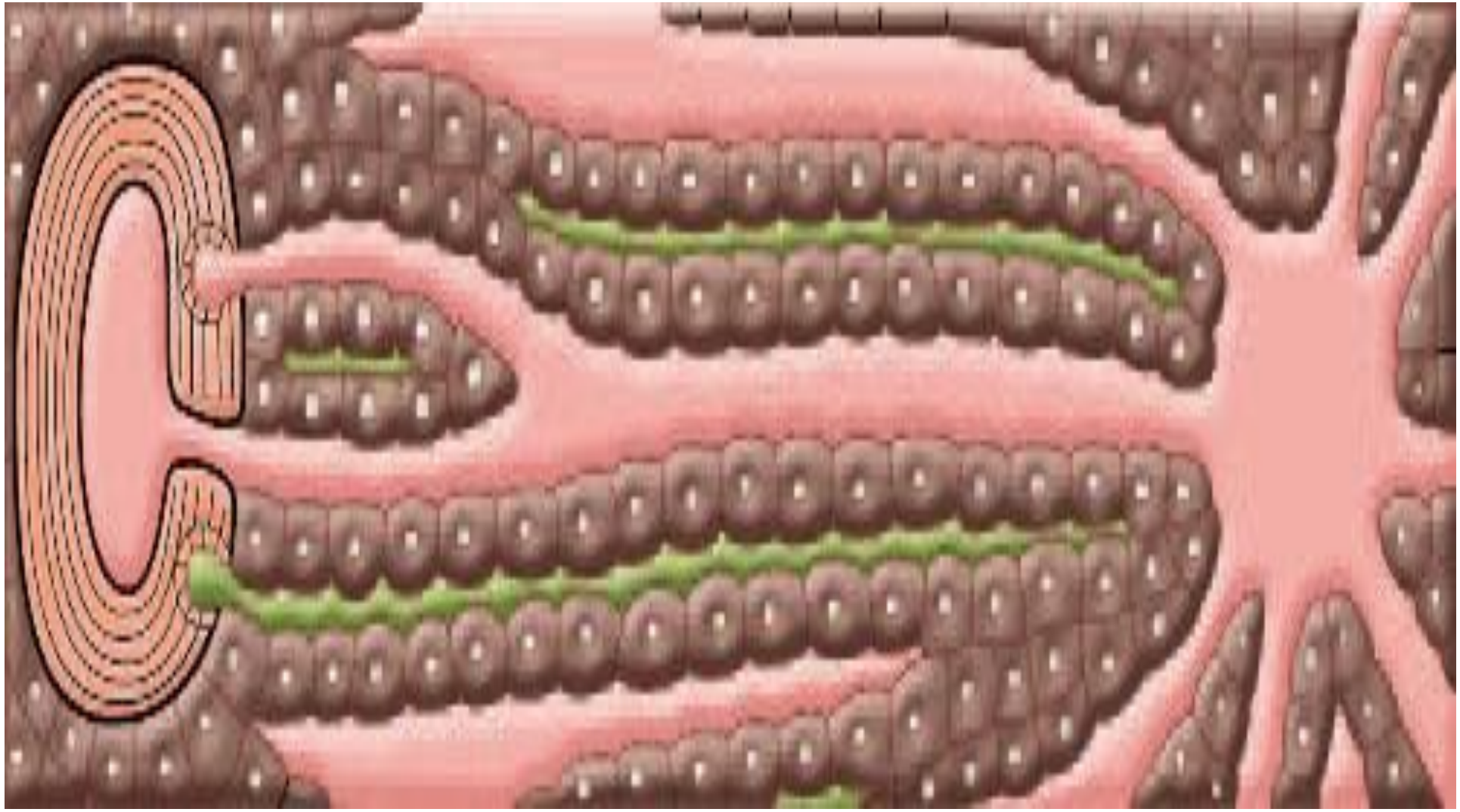
③ Secretory & excretory functions *(Formation of* bile*)*

- **Volume: 500 - 1500 ml/day**
- **PH: Alkaline**
- **Composition**
  - ❶ Water 97%
  - ❷ **Bile salts & bile pigments**
  - ❸ Cholesterol ,fatty acids, lecithin and fat
  - ❹ Inorganic salts
  - ❺ Alkaline phosphates

NB: All are dissolved in **alkaline** electrolyte

- Secretion: Bile is secreted by liver cells into bile duct which drains into duodenum
- Bile pigments
  - Is a greenish yellow pigment
  - Is metabolic **end product** of hemoglobin
  - Is excreted mostly in stool [gives stool its brown color]

Bile canaliculi



- Bilirubin metabolism & excretion:

In reticuloendothelial system (RES)

The old red cells are destroyed in: Tissue macrophage system

- **Hemoglobin:** Splits into:

  1-Globin which is split into amino acids

  2-Haem which is split into:

  1. Iron: Which is reused for hemoglobin synthesis
  2. Chain of 4 pyrole nuclei (rings): Which transformed to biliverdin (green) which is reduced to bilirubin (yellow), then released to plasma

- **Hemoglobin:** Splits into:

  1-Globin which is split into amino acids

  2-Haem which is split into:

  1. Iron: Which is reused for hemoglobin synthesis
  2. Chain of 4 pyrole nuclei (rings): Which transformed to biliverdin (green) which is reduced to bilirubin (yellow), then released to plasma

- **In liver:**
  - Free bilirubin without albumin is **uptaken** by the liver cells by an **active** process (active re-uptake)
  - Then, bilirubin is bound to cytoplasmic proteins
  - In smooth endoplasmic reticulum: one bilirubin molecule is **conjugated** to 2 glucuronic acid molecules in a reaction catalysed by the **glucuronyl transferase enzyme** forming →cholebilirubin [or conjugated or direct bilirubin] which is more water soluble than the free bilirubin
  - Glucuronyl transferase is:

- Activated by:	- Inhibited by:
1- Barbiturate	1- Chloramphenicol
2- Anticonvulsant	2- Pregnandlol
3- Antihistaminic	

  - Conjugated bilirubin is then excreted by: An **active** transport process into the bile canaliculi

- **In the intestine:**

Most of conjugated bilirubin passes via bile duct to intestine. (intestinal mucosa is relatively impermeable to conjugated bilirubin)

❶ **50% of conjugated bilirubin**: Is reduced by bacterial action to urobilinogen (highly soluble)

❷ **The rest of conjugated bilirubin**: Is deconjugated in intestine and is absorbed into portal circulation then excreted again by the liver (enterohepatic circulation)

Fate of urobilinogen:

Most	Some	5%
1. Most of it is changed into **sterocobilinogen** & oxidized to **sterocobilin** [which is responsible for brown color of stool] & excreted in stool	1. Some is absorbed into portal vein [via enterohepatic circulation] to be excreted by liver back into gut	1. 5% enters systemic circulation & excreted in urine urobilinogen NB: Urobilinogen is oxidized **in air** to urobilin

NB: Plasma bilirubin is 2 types:

① Hembilirubin

= [Unconjugated bilirubin]

= [Indirect bilirubin]

= [Free bilirubin]

It is the main form

It gives indirect Van den Bergh reaction

② Cholebilirubin

= [Conjugated bilirubin]

= [Direct bilirubin]

It is low concentration

It gives direct Van den Bergh reaction

What is meant by Van den Berg reaction?

- It means adding Erlich reagent to plasma
- When plasma contains **conjugated** bilirubin it gives **directly** pink

What is meant by Van den Berg reaction?

- It means adding Erlich reagent to plasma
- When plasma contains **conjugated** bilirubin it gives **directly** pink color
- When plasma contains **unconjugated** bilirubin, it gives red color **after** addition of alcohol

JAUNDICE

JAUNDICE

Jaundice (Icterus)

★ **Definition:** It is yellowish tinge of the skin, mucous membrane & sclera due to hyperbilirubinaemia above 2 mg% (Normally Bilirubin level is up to 0.5 mg%)

★ **Causes of hyperbilirubinaemia:**

1-Increased free bilirubin is caused by:

1. Excess production eg hemolytic anemia
2. Decreased uptake of bilirubin into hepatic cells
3. Disturbed intracellular protein binding or conjugation

2-Increased conjugated bilirubin is caused by:

1. Disturbed secretion of conjugated bilirubin into bile canaliculi
2. Entrahepatic or extrahepatic duct obstruction

★ Types of Jaundice:

❶ Hemolytic Jaundice *(Prehepatic = Retention)*

-Causes: Due to excessive hemolysis of RBCs eg:

[A] Hemolytic anemia as:
1-Spherocytosis
2-Sickle cell anemia
3-Thalassemia
4-Erythroblastosis fetails
5-Malaria

[B] Incompatible blood transfusion

❷ Hepatic Jaundice *(Hepatocellular)*

- Causes: Liver diseases: Damage of liver cells by:
  1- Viral hepatitis
  2- Drugs eg tetracycline & chloramphenicol
  3- Chemicals eg chloroform, arsenic & mercury

[1] The liver is: Unable to uptake haembllirubin
[so haembilirubin increases in blood]

[2] The liver is: Unable to excrete cholebilirubin in intestine
[so, cholebilirubin & bile salts are regurgitated to the blood & excreted in urine]

[3] Decrease formation of urobilinogen in urine

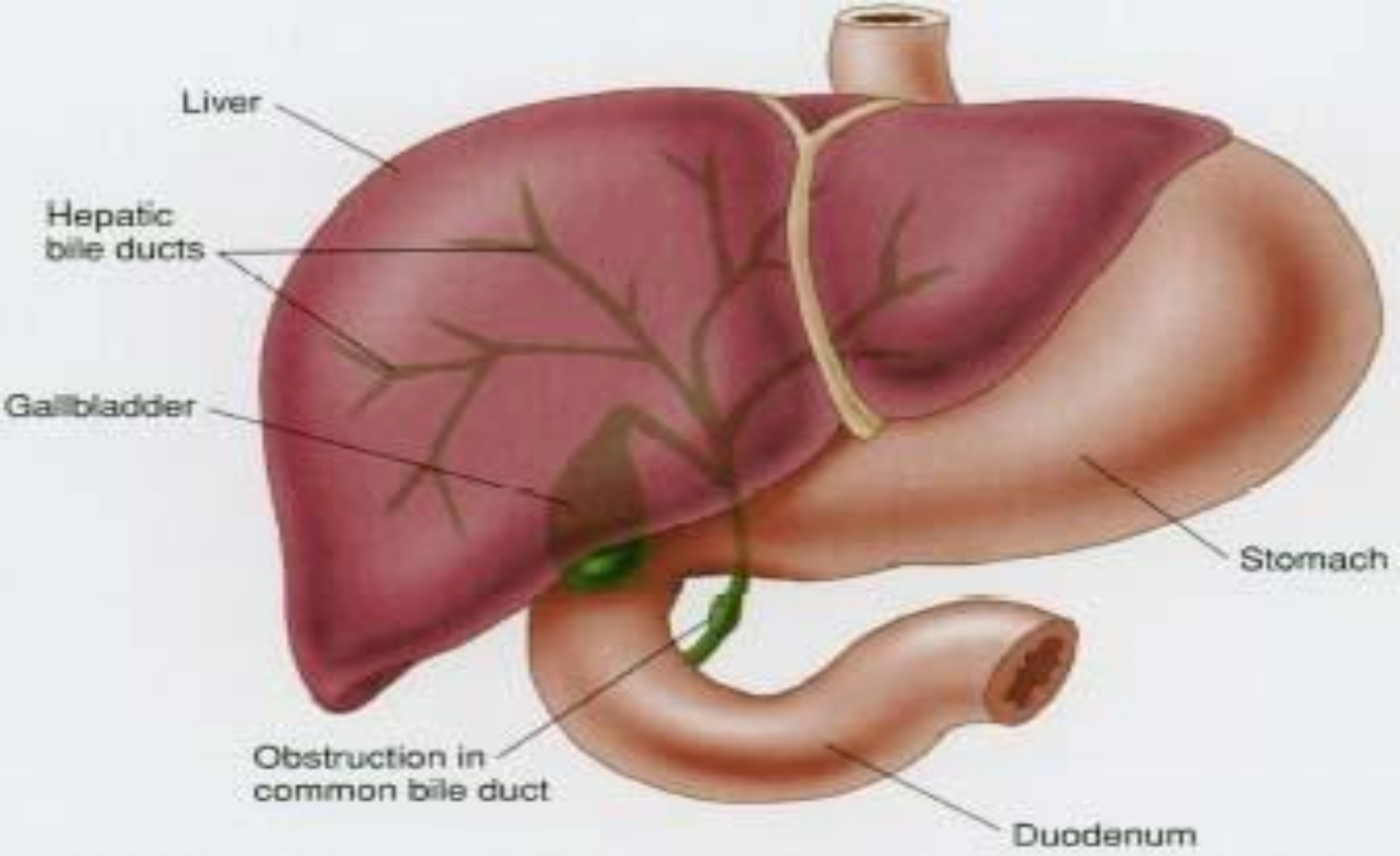
[4] Decrease formation of sterocobilinogen in stools

- There is impairment of liver functions

❸ Cholestatic Jaundice *(Posthepatic = Regurgitation = Obstructive Jaundice)*

- Causes: Due to obstruction of bile duct:
  - 1-Intrahepatic:
    - Hepatitis
    - Cirrhosis
  - 2-Extrahepatic:
    - Cancer head of pancreas
    - Bile duct stone
- There is: Inability of cholebillrubin (conjugated bilirubin) & bile salts to reach the intestine, so they **regurgitate** back to the blood either:
  - 1-Directly: Via the liver sinusoids or
  - 2-Indirectly: By absorption from bile ducts & lymphatics
- There is increase cholebilirubin in blood then they pass in urine
- Urine is very dark (due to cholebilirubin): So, it is Coca-Cola or Tea like color
- Stool is:
  - Pale (no sterocobilinogen)
  - Steatorrhea (bulky and greasy)
- Blood: There is increased cholesterol and **alkaline phosphatase** levels
- Presence of bile salts in blood lead to:
  - ① Pruritis (itching)
  - ② Bradycardia

Obstructive Jaundice



Types of jaundice	Pre hepatic	Hepatic	Post hepatic
Type of bilirubin elevated	unconjugated bilirubin	Both conjugated & unconjugated bilirubin	conjugated bilirubin
Serum bilirubin - Van den Bergh test	Indirect positive	Biphasic	Direct positive
Urine ✓ Conj.Bilirubin ✓ Urobilinogen ✓ Bile salt	 Absent +++ Absent	 ++ + early, obst.-dec. +	 +++ Absent ++
Urine color	Normal-**Acholuric**	Dark - **Choluric**	Dark - **Choluric**
Stool color	Dark brown colour	N /decreased	Clay colored stools
AST & ALT	Normal	Very high	inc.
ALP Levels	Normal	2-3 times increased	10-12 times inc.

Thank You